Forest Economics

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Economics (A08002)

Short Title: Forest Economics
Faculty Science and Computing
Department Science
Cluster Forestry
Author MSOTTOCORNOLA
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module will explore the role of economics in shaping forest management. The module will focus on developing financial appraisal techniques to aid sound management decision making. The student will gain the professional competence in analysing forestry as an investment, comparing rotations, species, treatments, and subjecting results to sensitivity analysis.

Programmes

`ECOS-0001`	BSc in Forestry (`WD_SFORS_D`)	3 / 5 / M

Indicative Content

- Principles of economics, of forestry macroeconomics and its contribution to the national economy, and of budgeting
- Time scale, compounding and discounting, annuities; interest rate, discount rate and inflation rate
- Forest costs and revenues: timber sales, price-size relationships, grant aid, land rental and sales, hunting and sports activities, sites, masts, wind farms; scheduling costs and revenues on a spreadsheet; presenting values of costs and revenues; calculating present values in Excel
- Net discounted revenue; net present value; calculating NPV & NDR in Excel; soil expectation values; internal rate of return; Calculating IRR in Excel; forestland purchase price and evaluation; open space value in forestry valuation; forest investment analysis project
- Operations costs and benefits, ground preparation; applying fertiliser & herbicide; pruning and thinning; sensitivity analysis
- Rotations, estimating financial rotation, effect of windthrow on financial rotation, estimating financial consequences of fire or forest reconstitution

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise forest economics and forestry's contribution to the national economy.
2. Complete analysis of present value, net present revenue, internal rate of return and soil expectation value of a forest.
3. Use financial appraisal techniques to aid forest management decision making.
4. Analyse the likely risks, returns and taxation implications when investing in Irish forestry.
5. Complete a forest valuation and a forest budget.
6. Use computer software programme to calculate the financial implications of forest management decisions.

Learning and Teaching Methods

- Lectures: will be used to teach the economic theory and forest sector conditions that underpin financial analysis of forestry as an investment.

- Computer practicals: will allow students to develop financial appraisal tools using spreadsheets; spreadsheet based examples will facilitate demonstrating the effect of key parameters, such as discount rate, time, forest productivity etc. on the decision support tool.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Assignment	50%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "GrowFor - Forest Planning Models." http://www.coford.ie/toolsservices/growfor/
- Bright, G. _Forestry Budgets and Accounts_. UK: CAB International, 2001.
- Christie, J.M. and P.N. Edwards. _Yield Models for Forest Management_. Edinburgh, UK: Forestry Commission, 1991.
- Clinch, J.P. _Economics of Irish Forestry_. Ireland: COFORD, 1999.

Supplementary Material

- Department of Agriculture, Food and the Marine. _Code of Practice - Valuation of Commercial Forest Plantations. Draft_ by Little, D., J. Phelan, T. McDonald and H. Philips.. Ireland. 2012.
- Fraser, T., G.P. Horgan and G.R. Watt. _Valuing forests and forest land in New Zealand: Principles and practice_ . New Zealand: Rotorua : Forest Research Institute, 1985.
- Hart, C. _Practical forestry for the agent and surveyor_. 3rd ed. USA: Alan Sutton Publishing Inc., 1991.
- Helliwell, D.R. _Economics of Woodland Management_. UK: Chichester: Packard, 1988.
- Philips, H. "Approaches to forestry investment." _Irish Forestry_ 56, 1. (1999): 2-11.
- Williams, M.R.W. _Decision-making in forest management_. USA: Research Studies Press, Wiley, 1988.

Requested Resources

- Room Type: Computer Lab